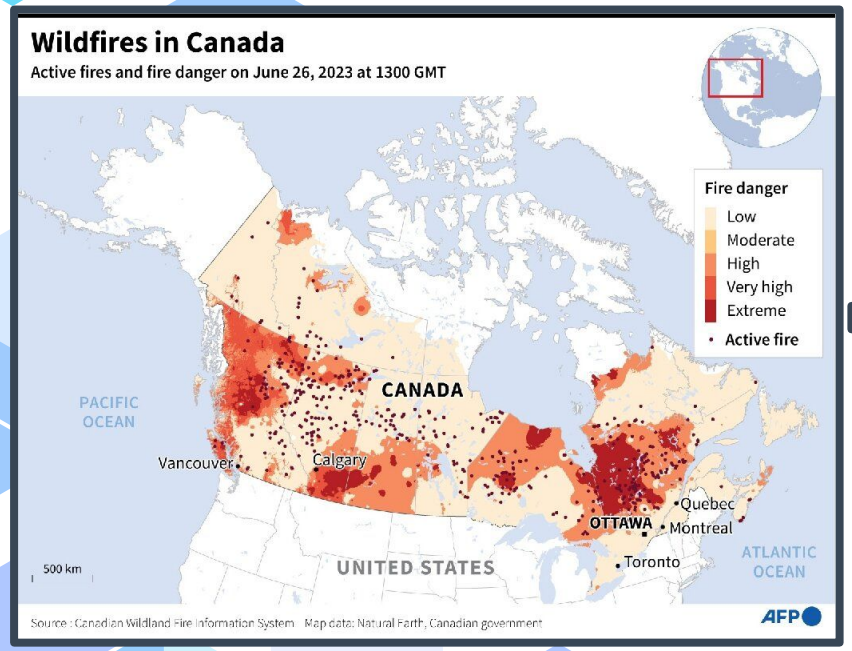
HealthyCleanAir4All

Data Analysis

Ganeshan Venu & Armaan Sarkhel

An Overview


Wildfires in Canada
Active fires and fire danger on June 26, 2023 at 1300 GMT
Fire danger
Low
Moderate
High
Very high
Extreme
Active fire
CANADA
PACIFIC OCEAN
Vancouver
Calgary
OTTAWA
Quebec
Montreal
Toronto
UNITED STATES
ATLANTIC OCEAN
500 km
Source : Canadian Wildland Fire Information System Map data: Natural Earth, Canadian government
AFP

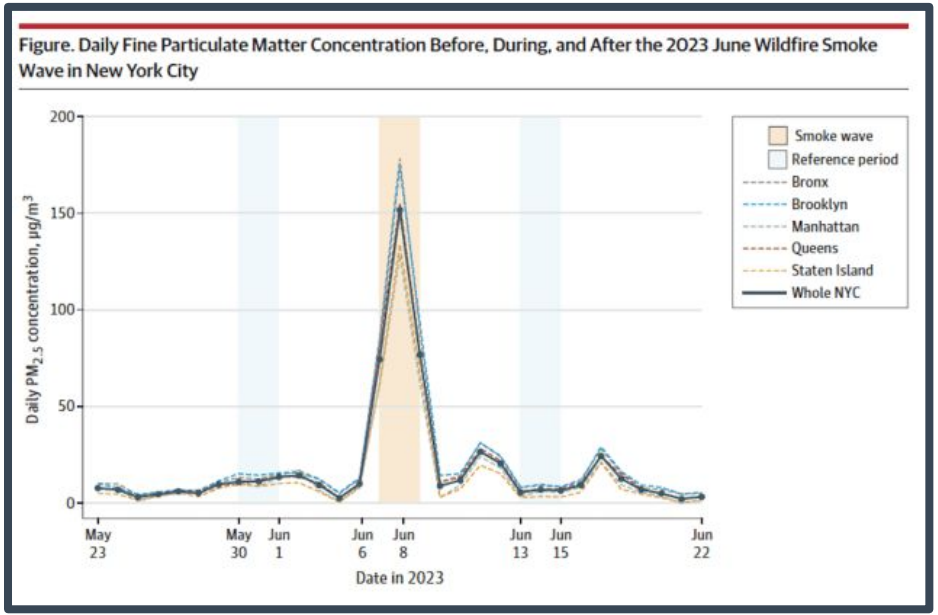
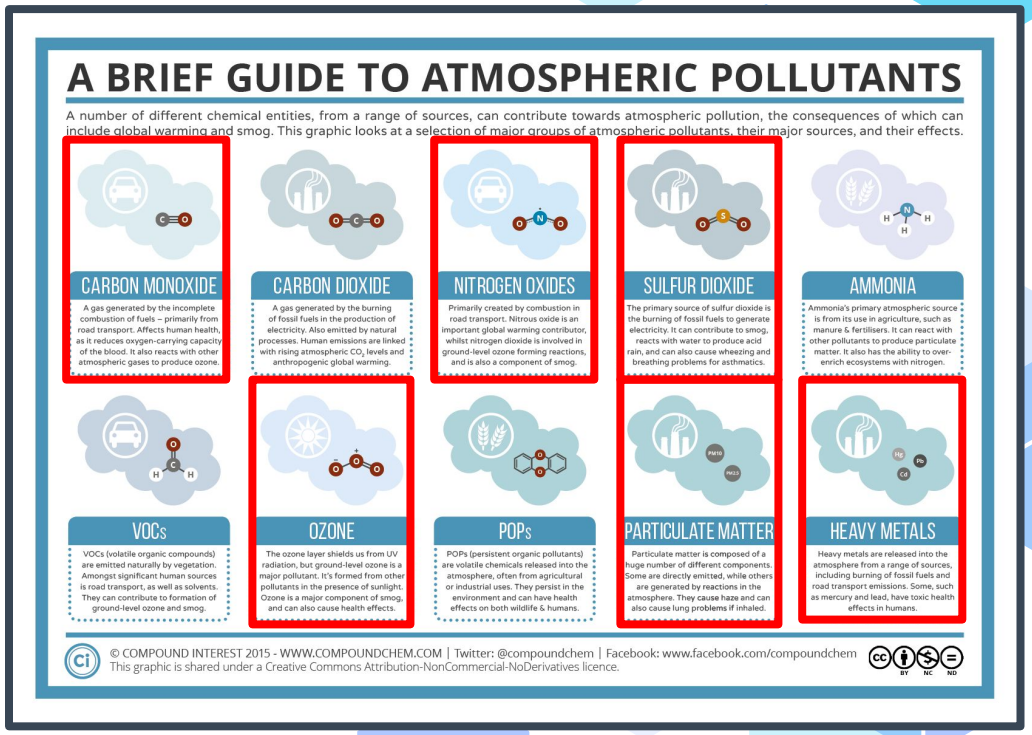


Figure. Daily Fine Particulate Matter Concentration Before, During, and After the 2023 June Wildfire Smoke Wave in New York City
Daily $PM_{2.5}$ concentration, $\mu g/m^3$
200
150
100
50
0
May 23
May 30
Jun 1
Jun 6
Jun 8
Jun 13
Jun 15
Jun 22
Date in 2023
Smoke wave
Reference period
Bronx
Brooklyn
Manhattan
Queens
Staten Island
Whole NYC

What’s Air Quality?

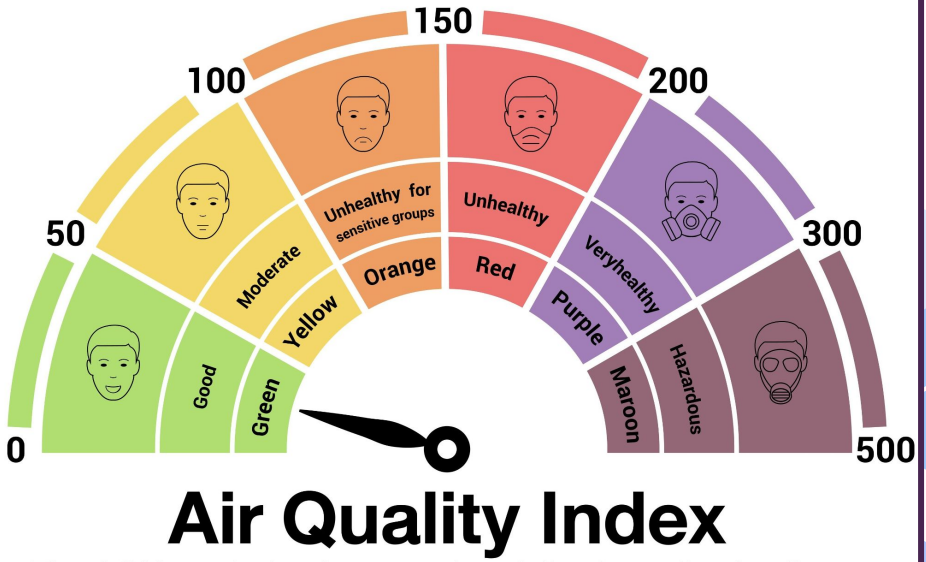
Air Quality

- **Air quality is a measure of the main pollutants in the air we breathe**
  - Particulate Matter (PM)
  - Ozone
  - Sulfur Dioxides
  - Nitrogen Oxides
  - Lead
  - Carbon Monoxide

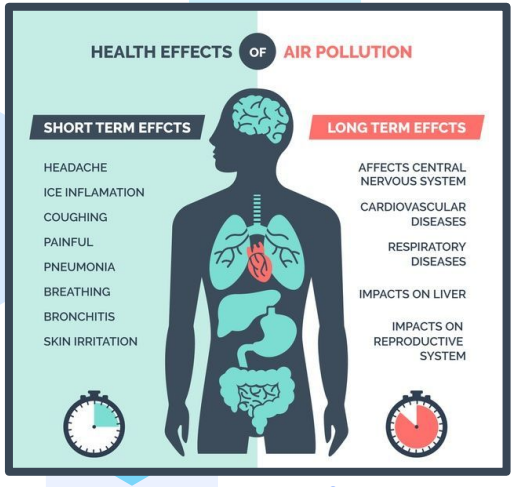


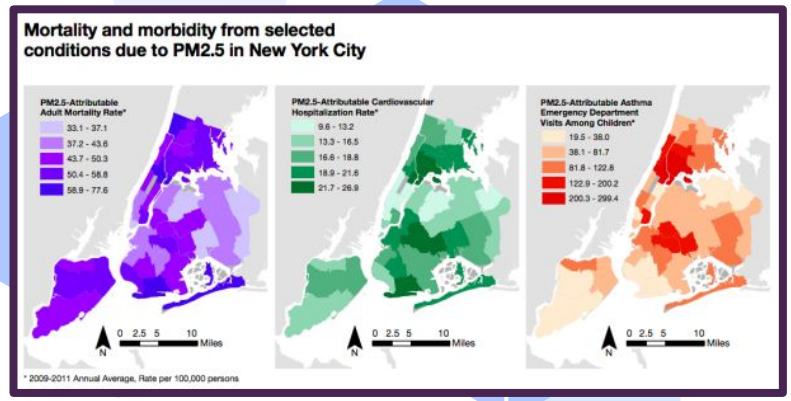
Air Quality Index

Measures air pollution on a standardized scale from 0 (good) to 500 (hazardous).



Potential Hazards





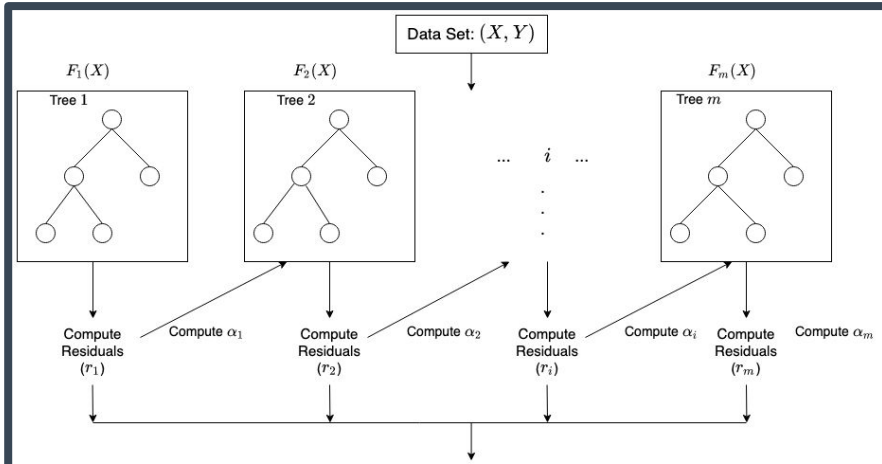
- **Negative Effects**
  - Asthma & Respiratory Conditions
  - Linked to heart and lung diseases
- **Environmental Damage**
  - Acidification
  - Eutrophication

Analysis,

Predictions & More!

XGBoost

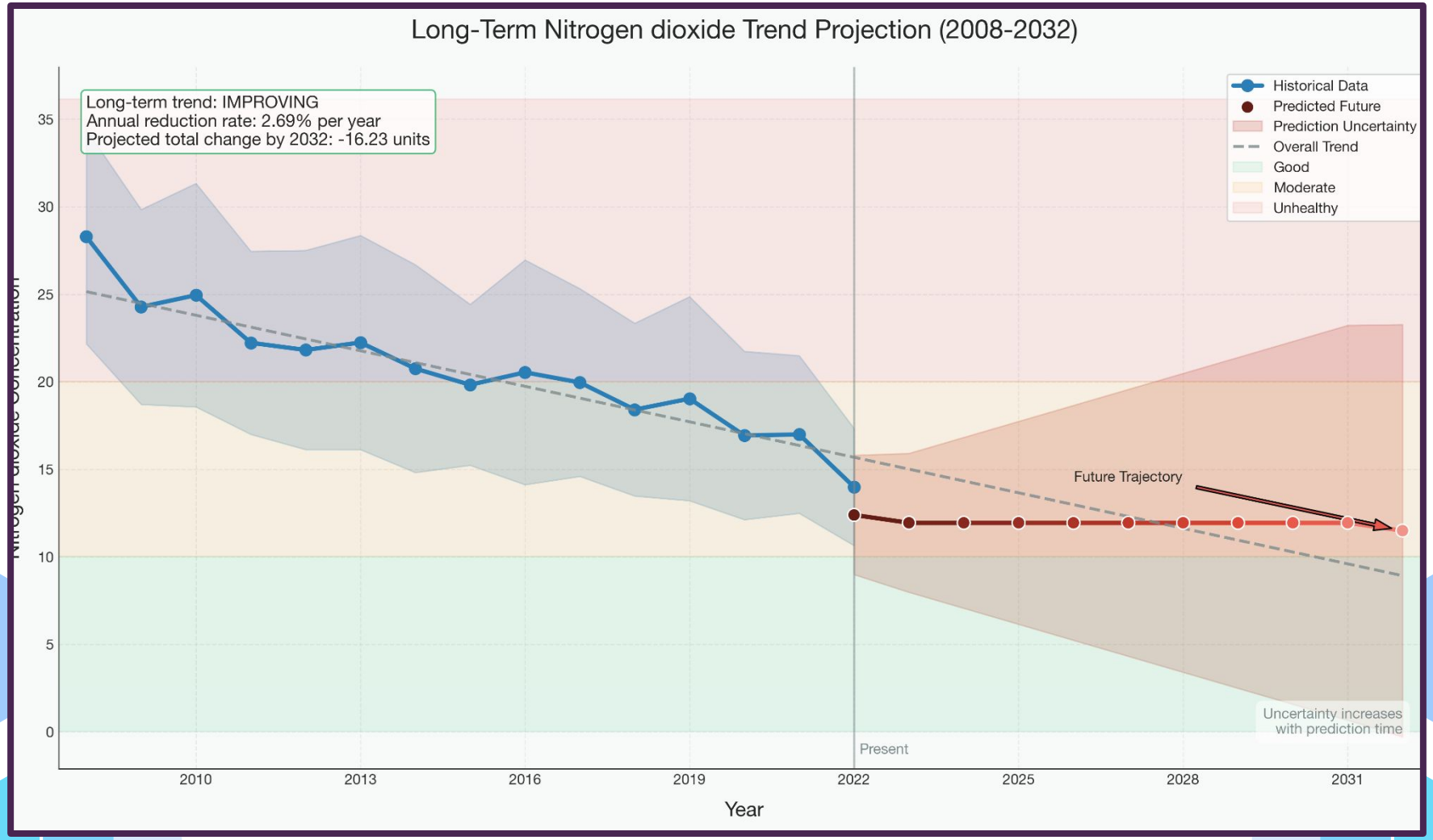
- "Extreme Gradient Boosting"
- Advanced Machine Learning Algorithm
- Utilizes Gradient Boosting & Weighted Voting
- Can derive relationships between data



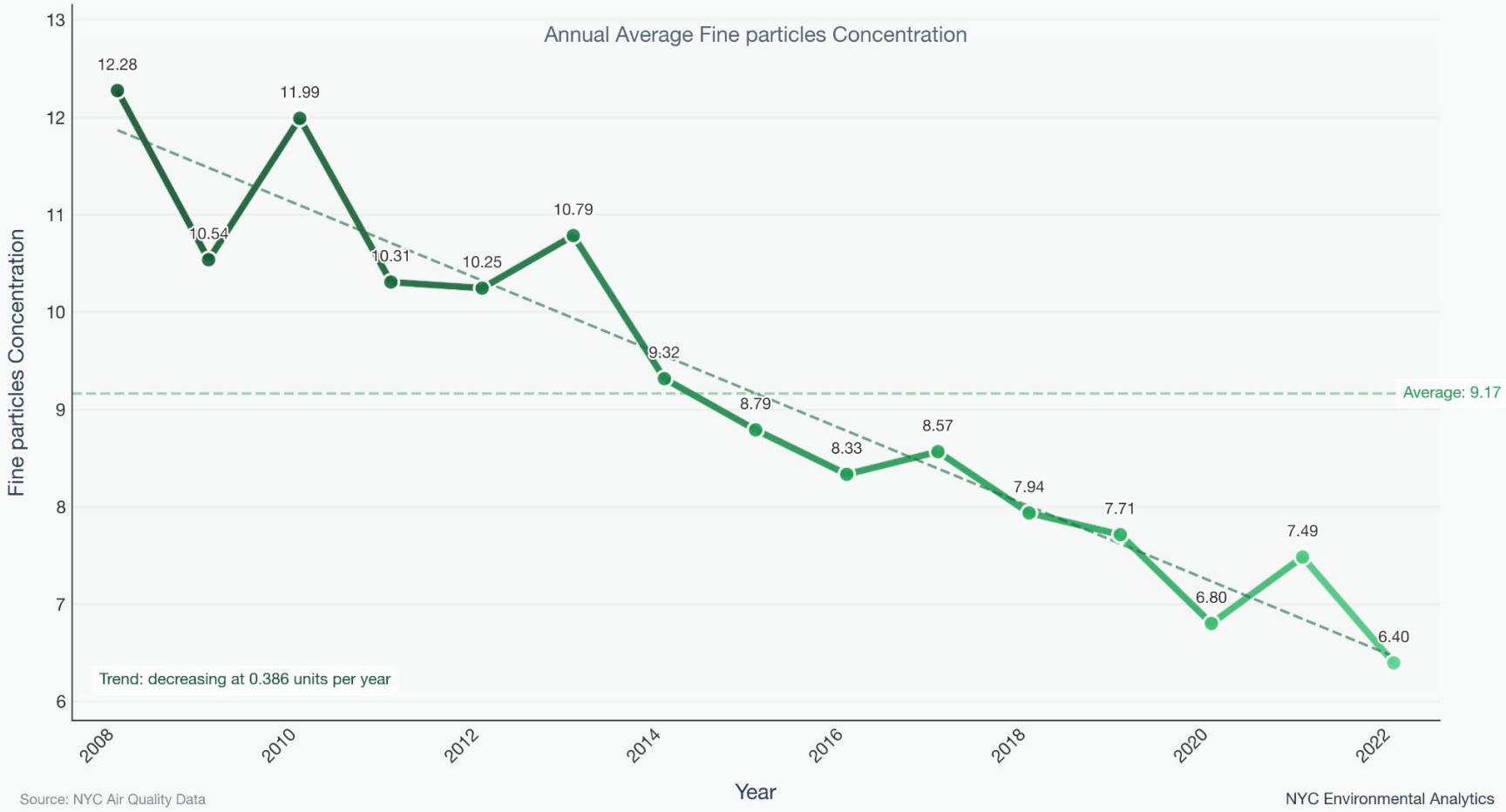
$$F_m(X) = F_{m-1}(X) + \alpha_m h_m(X, r_{m-1}),$$

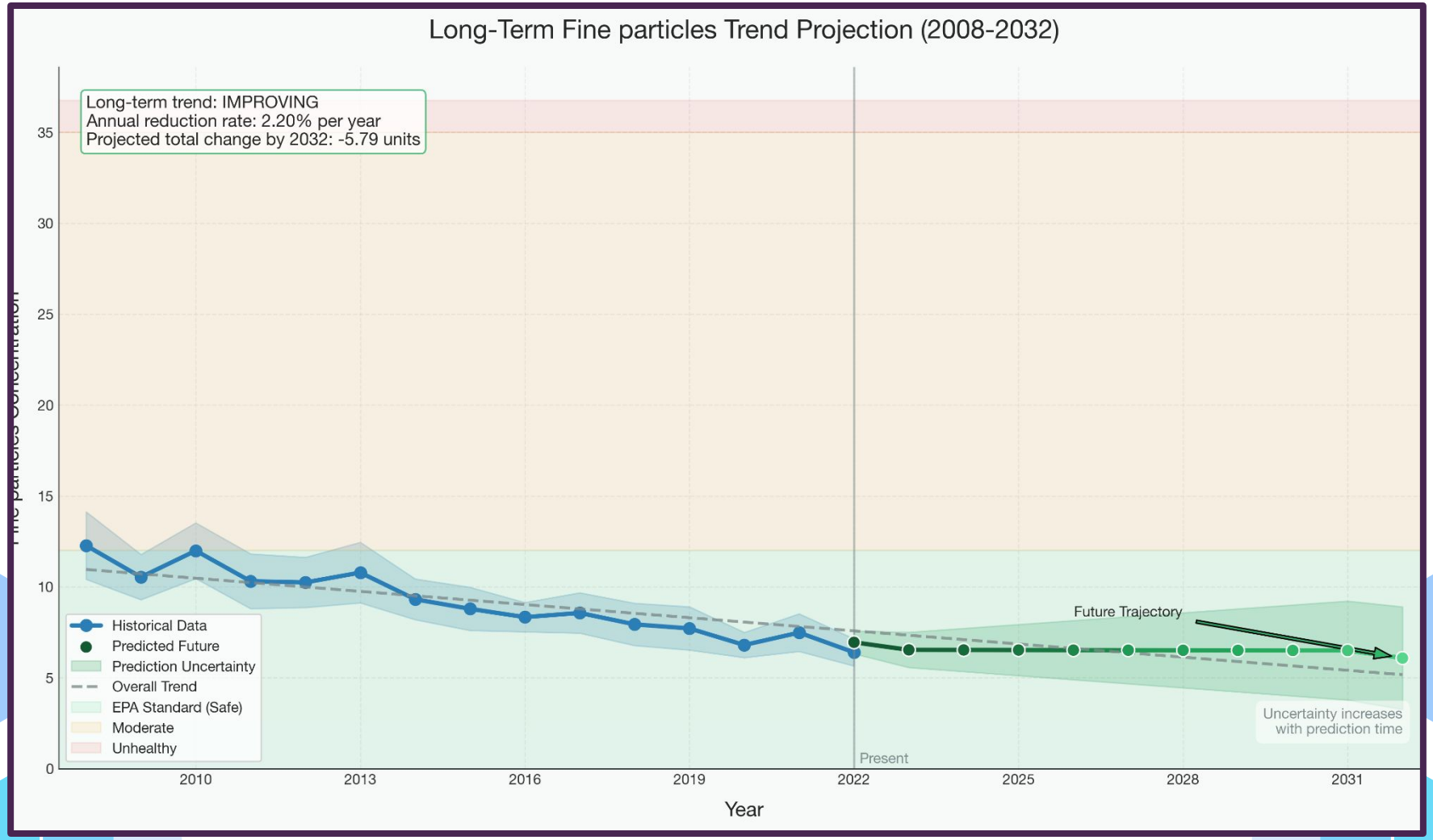
where α_i, and r_i are the regularization parameters and residuals computed with the ith tree respectfully, and h_i is a function that is trained to predict residuals, r_i using X for the ith tree. To compute α_i we use the residuals computed, r_i and compute the following: $arg \min_{\alpha} = \sum_{i=1}^{m} L(Y_i, F_{i-1}(X_i) + \alpha h_i(X_i, r_{i-1}))$ where $L(Y, F(X))$ is a differentiable loss function.

NYC Annual Nitrogen dioxide Index Trends


Long-Term Nitrogen dioxide Trend Projection (2008-2032)
Long-term trend: IMPROVING
Annual reduction rate: 2.69% per year
Projected total change by 2032: -16.23 units
Historical Data
Predicted Future
Prediction Uncertainty
Overall Trend
Good
Moderate
Unhealthy
Future Trajectory
Uncertainty increases with prediction time
Present
Nitrogen dioxide Concentration
Year
35
30
25
20
15
10
5
0
2010
2013
2016
2019
2022
2025
2028
2031

NYC Annual Fine particles Index Trends




Long-Term Fine particles Trend Projection (2008-2032)
Long-term trend: IMPROVING
Annual reduction rate: 2.20% per year
Projected total change by 2032: -5.79 units
Fine particles Concentration
35
30
25
20
15
10
5
0
Historical Data
Predicted Future
Prediction Uncertainty
Overall Trend
EPA Standard (Safe)
Moderate
Unhealthy
Future Trajectory
Uncertainty increases with prediction time
Present
2010
2013
2016
2019
2022
2025
2028
2031
Year

High Efficiency Air Filtration

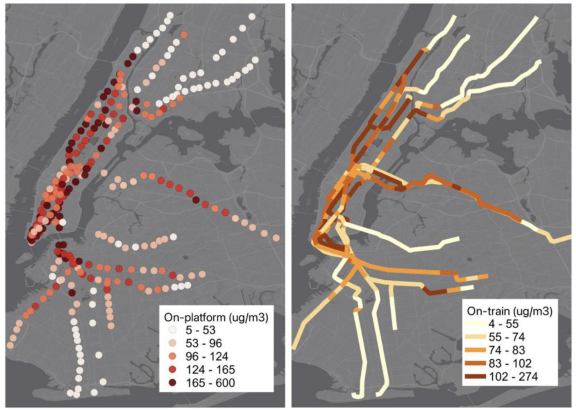


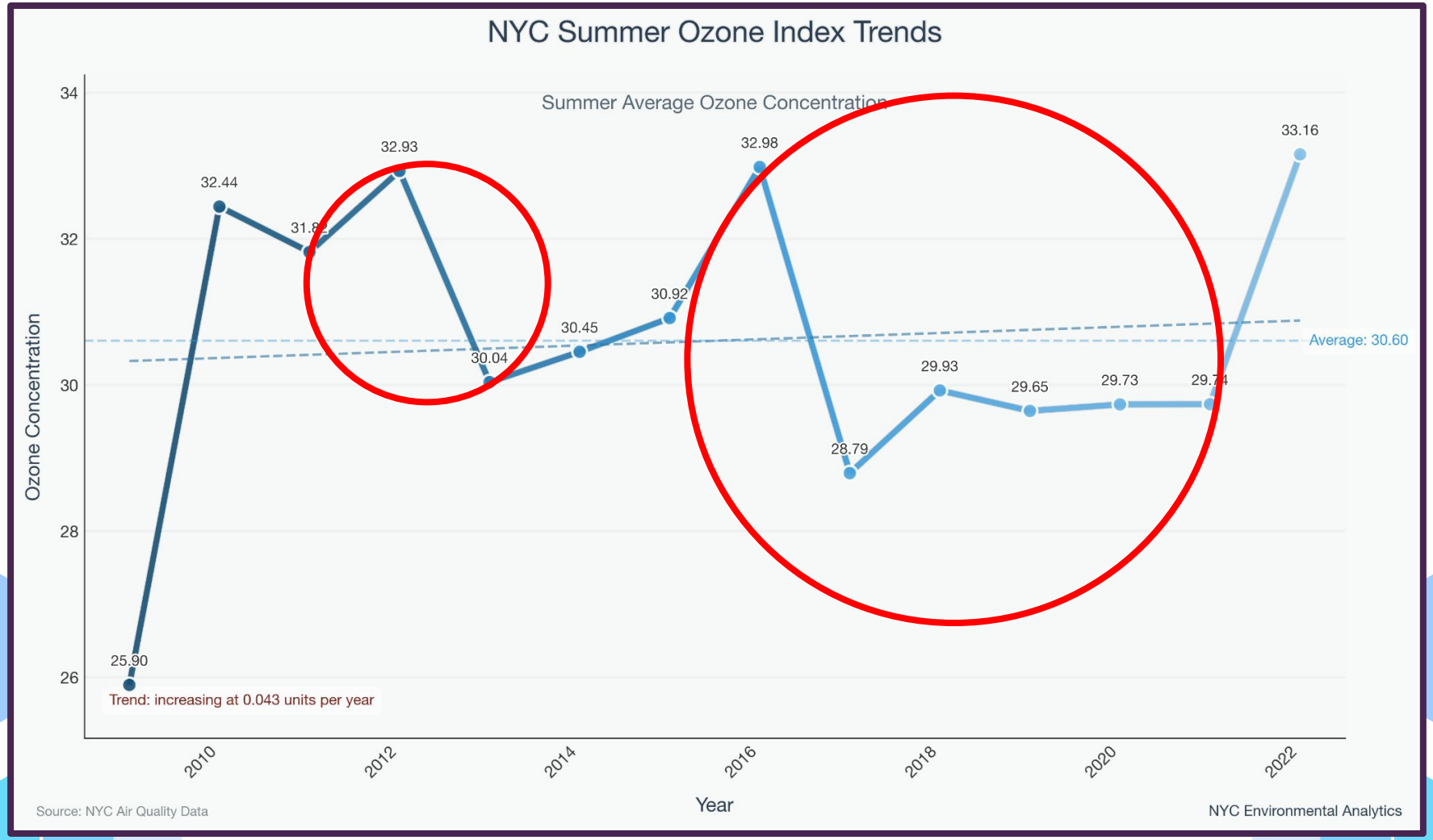
Figure 2:
(left) Average PM2.5 concentration on station platforms and (right) inside train cars between stations. (right).

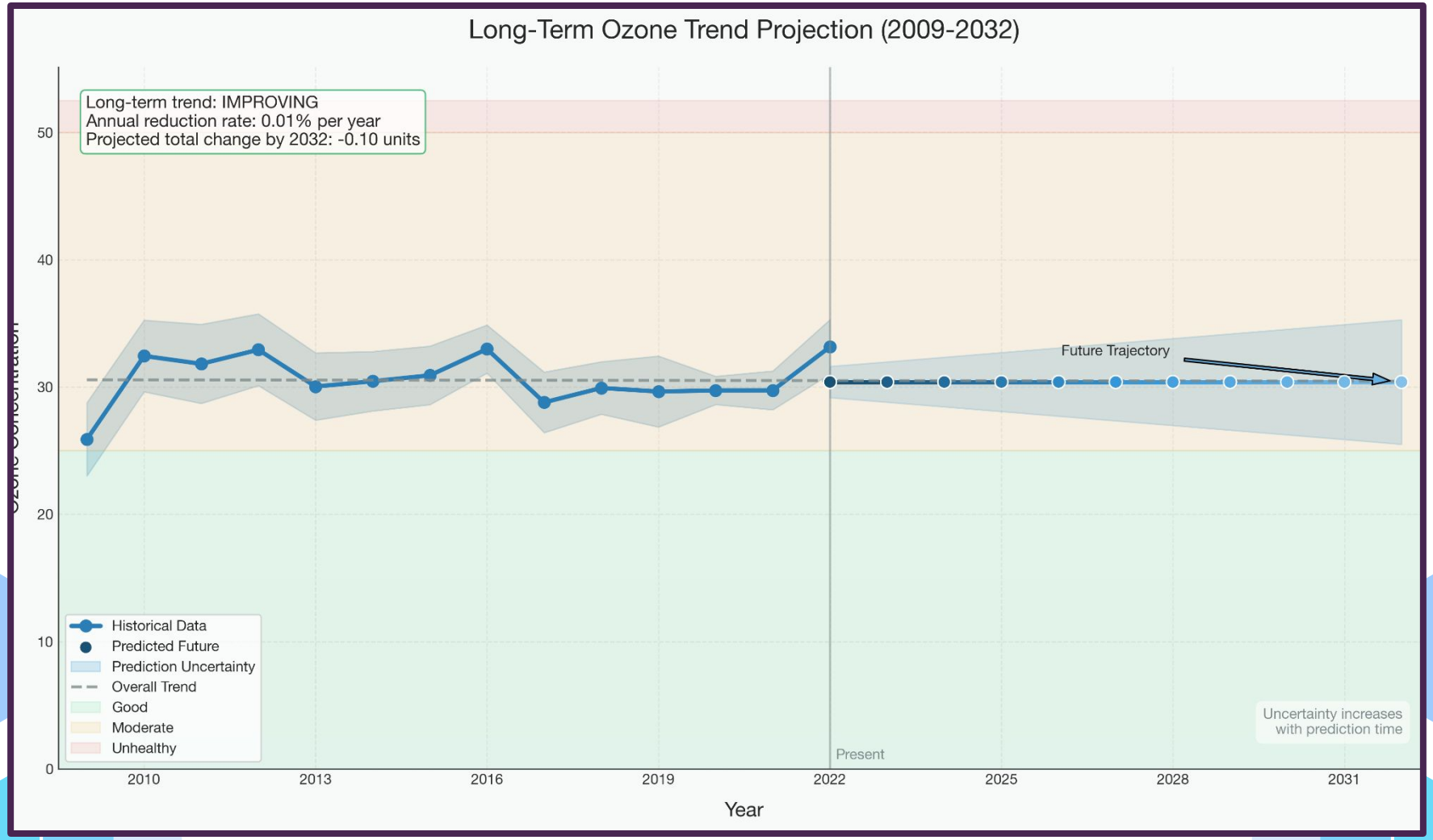
Install HEPA and activated carbon filters in subway stations, bus stops, and public buildings

Table 3.4: HEFA activated carbon filters implementation

Country	Impacts	Source
China	China has been grappling with severe air pollution issues, particularly in cities like Beijing and Shanghai. Studies have shown that the use of air purifiers with HEPA and activated carbon filters has led to improvements in indoor air quality in China, reducing exposure to fine particulate matter (PM2.5) and volatile organic compounds (VOCs).	(Li, 2017)
USA	Activated charcoal filters in vehicle cabin air filters can help reduce exposure to air pollutants while driving. A study published in Environmental Science and Pollution Research found that activated carbon filters in car air conditioning systems effectively reduced in-cabin concentrations of volatile organic compounds (VOCs) and particulate matter (PM)	US EPA 2023

One such type of bus can be found in the United Kingdom, where the Go-Ahead Group, one of the UK's largest bus and rail operators, has launched the Bluestar Bus, an air-filtering bus that cleans the air as it plies its daily route. The Bluestar Bus is capable of removing fine particulate matter in the atmosphere with an efficiency of up to 99.5 percent, effectively sucking up dirty air and releasing it as clean, purified air that's safe to breathe.


NYC Summer Ozone Index Trends
Summer Average Ozone Concentration
Ozone Concentration
34
32
30
28
26
25.90
32.44
31.82
32.93
30.04
30.45
30.92
32.98
28.79
29.93
29.65
29.73
29.74
33.16
Average: 30.60
Trend: increasing at 0.043 units per year
2010
2012
2014
2016
2018
2020
2022
Year
Source: NYC Air Quality Data
NYC Environmental Analytics


Long-Term Ozone Trend Projection (2009-2032)
Long-term trend: IMPROVING
Annual reduction rate: 0.01% per year
Projected total change by 2032: -0.10 units
Future Trajectory
Historical Data
Predicted Future
Prediction Uncertainty
Overall Trend
Good
Moderate
Unhealthy
Uncertainty increases with prediction time
Present
0
10
20
30
40
50
2010
2013
2016
2019
2022
2025
2028
2031
Year

Dynamic AQI Congestion Fees

Greening Agency Operations

NEW

Reduce emissions from the DOT vehicle fleet

As part of its NYC Clean Fleet initiative, the City is creating the largest municipal electrical vehicle fleet in the United States. DOT current operates 639 sedans, of which 53 are plug-in hybrids or fully electric vehicles. The agency plans to replace 50 percent of the sedans retired each year with plug-in hybrids or fully electric vehicles. The City is also exploring strategies to green the agency's light and heavy-duty truck fleets.

Implement dynamic congestion charges based on hazardous AQI levels to cut traffic and emissions.

Taxi of Tomorrow Given the Green Light for New York City

September 20, 2012

2013-04-22

Nissan and New York City launch LEAF EV Taxi Pilot

Position papers

Mayor de Blasio Announces That Citi Bike Rides Surged 40 Percent in 2016 -- Nearly 14 Million Trips Taken

December 29, 2016

New York City's bike share system shatters annual ridership record for third consecutive year, exceeding 2015 record by 4 million trips

“Preserve and cherish the pale blue dot, the only home we’ve known”

—Carl Sagan

Thank you!

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